

capital and diversify risk. The next major indicator is market liquidity, which is measured by value-traded ratio and turnover ratio. Value traded ratio equals the total value of traded shares in the stock market divided by GDP. While, in turn, turnover ratio is the value of total shares traded divided by market capitalisation. The next important indicator of stock market development is the volatility parameter, which conceptualises the asset price movement in a stock market and conveys important signals for its development.

The average value of market capitalisation ratio (MCR) for India from 2003 to 2019 was 76.26 per cent. The minimum MCR was 45.93 per cent in 2003 and the maximum was 149.51 per cent in 2007. In the latter part of 2007, the US subprime lending crisis broke out and in no time it became a global economic crisis. Due to this, the MCR fell drastically to 53.98 per cent. In the wake of demonetisation undertaken by the Central Government on 8th Nov 2016, MCR declined to 68.27 per cent from 72.08 per cent in 2015. Moreover, with the advent of Goods and Service Tax (GST) on 1st July 2017, the MCR increased to 87.89 per cent and then declined to 75.81 per cent in 2019 and continued on a declining trend subsequently because of COVID-19 pandemic. The stock market turn over ratio (TOR) was 68.25 per cent on an average from 2003 to 2019. In 2003, the TOR was 94.47 per cent. It had grown to 142.99 per cent during the global financial crisis period in 2008. However, in the demonetisation period of 2016, it declined to 51.18 per cent and further to 50.87 per cent during the GST period in 2017. It increased to 58.07 per cent in the post GST period in 2018 and then declined to 28.89 per cent in 2019 and continued the trend during COVID-19 pandemic period. The average value of the stock market value traded ratio (VTR) was 51.96 per cent with a minimum of 28.96 per cent in 2013 and a maximum of 93.97 per cent in 2007. It declined to 77.19 per cent in 2008 on account of the global financial crisis and further to 34.94 per cent in 2016 because of demonetisation before recovering to 44.71 per cent in 2017. Finally, the average value of the stock price volatility from 2003 to 2017 was 21.83 per cent. The minimum was 12.76 per cent in 2017 and a maximum of 43.74 per cent in 2009.

10.4 STRUCTURE AND PERFORMANCE OF INDIAN STOCK MARKET

There are 23 stock exchanges in India, the first being the Bombay Stock Exchange (BSE), which began formal trading in 1875, making it one of the oldest in Asia. In 2010, the number of stock exchanges in India reduced to 19 as the Magadh, Mangalore, Hyderabad and Saurashtra Kutch stock exchanges have been derecognised by the market watchdog SEBI. As of Jan 17, 2020 the total number of recognised stock exchanges in India has reduced to 9 (see Table 10.1) as the Hyderabad Securities and Enterprises Ltd (erstwhile Hyderabad Stock Exchange), Coimbatore Stock Exchange Ltd, Saurashtra Kutch Stock Exchange Ltd, Mangalore Stock Exchange, Inter-Connected Stock Exchange of India Ltd, Cochin Stock Exchange Ltd, Bangalore Stock Exchange Ltd, Ludhiana Stock exchange Ltd, Gauhati Stock Exchange Ltd, Bhubaneswar Stock Exchange Ltd, Jaipur Stock Exchange Ltd, OTC Exchange of India, Pune Stock Exchange Ltd, Madras Stock Exchange Ltd, U.P. Stock Exchange Ltd, Madhya Pradesh Stock Exchange Ltd, Vadodara Stock Exchange Ltd, Delhi Stock Exchange Ltd and Ahmedabad Stock Exchange Ltd have been granted exit by SEBI.

Table 10.1: List of Stock Exchanges in India (As of Jan 2020)

S. No.	Name of the Recognised Stock Exchanges	Segments Permitted Valid upto	Recognition
1.	Bombay Stock Exchange Limited, Mumbai	a. Equity b. Equity Derivatives c. Currency Derivatives (including Interest Rate Derivatives) d. Commodity Derivatives e. Debt	Permanent
2.	Calcutta Stock Exchange Ltd		Permanent
3.	India International Exchange (India INX), Gujarat	a. Equity Derivatives (Equity Index Derivatives & Single Stock Derivatives) b. Commodity Derivatives c. Currency Derivatives d. Debt	28-Dec-20
4.	Indian Commodity Exchange Limited, Navi Mumbai	Commodity Derivatives	Permanent
5.	Metropolitan Stock Exchange of India Ltd., Mumbai	a. Equity b. Equity Derivatives c. Currency Derivatives (including Interest Rate Futures) d. Debt	15-Sep-20
6.	Multi Commodity Exchange of India Ltd., Mumbai	Commodity Derivatives	Permanent
7.	National Commodity & Derivatives Exchange Ltd., Mumbai	Commodity Derivatives	Permanent
8.	National Stock Exchange of India Ltd., Mumbai	a. Equity b. Equity Derivatives c. Currency Derivatives (including Interest Rate Derivatives) d. Commodity Derivatives e. Debt	Permanent
9.	NSE IFSC Ltd., Gujarat	a. Equity Derivatives (Equity Index Derivatives & Single Stock Derivatives) b. Currency Derivatives c. Commodity Derivatives d. Debt Securities (Masala Bonds)	Permanent

Source: SEBI**Note:** India International Exchange Ltd and NSE IFSC Ltd have Single Segment across all asset/product classes.

Before 1994, India's stock markets were dominated by BSE. In other parts of the country, the financial industry did not have equal access to markets and was unable to participate in forming prices, compared with market participants in Mumbai (Bombay). As a result, the prices in markets outside Mumbai were often different from prices in Mumbai. These pricing errors limited the order flow to these markets. Explicit nationwide connectivity and implicit movement toward one national market have changed this situation (Shah and Thomas, 1997). NSE has established satellite communications which give all trading members of NSE equal access to the market. Similarly, BSE and the Delhi Stock Exchange are both expanding the number of trading terminals located all over the country. The arbitrages are eliminating pricing discrepancies between markets.

Over the last few years, there has been a rapid change in the Indian securities market, especially in the secondary market. Advanced technology and online screen-based transactions have modernised the stock exchanges. In terms of the number of companies listed and total market capitalisation, the Indian equity market is considered large relative to the country's stage of economic development. Metropolitan Stock Exchange of India Limited (MSE) is recognised by the Securities and Exchange Board of India (SEBI) under Section 4 of Securities Contracts (Regulation) Act, 1956. The Exchange was notified as a "Recognised Stock Exchange" under Section 2(39) of the Companies Act, 1956 by Ministry of Corporate Affairs, Govt. of India, on December 21, 2012. MSE offers an electronic, transparent and hi-tech platform for trading in Capital Market, Futures and Options, Currency Derivatives and Debt Market segments. The Exchange has also received in-principle approval from SEBI for operationalising SME trading platform. MSE commenced operations in the Currency Derivatives (CD) Segment on October 7, 2008, under the regulatory framework of SEBI and Reserve Bank of India (RBI). MSE launched Capital Market Segment, Futures and Options Segment and flagship index 'SX40' on February 9, 2013, and commenced trading from February 11, 2013. Trading in the 'SX40' index derivatives began from May 15, 2013. 'SX40', is a free-float based index consisting of 40 large-caps, liquid stocks representing diverse sectors of the economy. The Debt Market Segment of MSE was launched on June 7, 2013, and trading commenced from June 10, 2013. The Exchange started live trading in cash-settled Interest Rate Futures (IRF), on Government of India security, in its Currency Derivative Segment from January 20, 2014.

On 9th Jan 2017, the Hon'ble Prime Minister of India, Shri Narendra Modi inaugurated India International Exchange Ltd. (IFSC), India's first international exchange. BSE is the first stock exchange in India to receive approval from SEBI on 26th September 2011 to launch Small and Medium Enterprises (SME) platform. The Platform facilitates capital raising by small and medium enterprises including start-up companies which are in their early stages of growth. It also provides easier entry and exit options for informed investors like angel investors, Venture Capital funds (VCFs) and Private Equities (PEs) etc. and equity financing which lowers the debt burden leading to lower financing cost and healthier balance sheet. SME IPO Index with a Base value of 100 as on 16th August 2012 is aimed at tracking the companies listed on the SME platform was launched on 14th December 2012. As on 31st August 2020, its value was 1,435.44.

BSE is the first stock exchange in India to launch startups platform on 22nd September 2018. The 'Startup Companies' seeking to be listed on the BSE Startups Platform should be in the sector of IT, ITES, Biotechnology and Life Science, 3D Printing, Space Technology, E-commerce, Hi-tech Defence, Drones, Nano Technologies, Artificial Intelligence, Big Data, Augmented/Virtual Reality, Egaming, Exoskeleton, Robotics, Holographic Technology, Genetic Engineering, Variable Computers Inside Body Computer Technology and other high-tech industries.

However, the Indian stock market is dominated by Bombay Stock Exchange and (BSE) and National Stock Exchange (NSE) which are occupying the 11th and 12th largest exchange in the World with the market capitalisation of 1.83 trillion and 1.41 trillion dollars respectively as of 2019. The average listed companies in Indian stock exchanges during the period from 1983 to 2018 was 4370 companies with a minimum of 1151 companies in 1983 and a maximum of 5999 companies in 1996. The latest number of companies in 2018 is 5065 companies. For comparison, the World average in 2018 based on 70 countries is 597 companies. A higher number of listed companies in Indian stock exchanges means that more companies use equity financing in their business. In February 2020, all the securities markets in the World is nosediving coupled with a precipitous fall in crude oil prices. India's market capitalisation to GDP ratio is 81 per cent compared to 51 per cent in Germany (Bloomberg). The market capitalisation of both BSE and NSE decreased by 6.2 per cent as on February 29, 2020, in comparison to previous month post to the first detection of COVID-19 patient in India on 30th Jan 2020 in the State of Kerala.

A snapshot of the growth rates, stock returns, volatility of stock prices, and FII net flows by comparing the major three events such as demonetisation, implementation of GST and the impact of COVID-19 is presented in Table 10.2. The demonetisation phase is considered from 9th November 2016 to 30th June 2017 and from 1st July 2017 to 29th January 2020 as treated as a post-GST phase. Similarly, the period starting from 30th January 2020 to 20 April 2020 are included in COVID-19 phase.

The growth rate of S&P BSE Sensex stock price shows a negative growth (–22.65 per cent) during COVID-19 pandemic phase as compared to positive growth rates of 12.07 per cent and 31.9 per cent during post demonetisation and GST phases, respectively. However, the other stock indices such as mid-cap, small-cap and all sectors except health care show drastic negative growth rates during COVID-19 phase as compared to demonetisation and GST phases. Both the returns and volatility series of stock indices reveal a precarious scenario during COVID-19 phase in comparison to other events.

All the stock indices except healthcare show negative returns and high volatility during this ongoing COVID-19 pandemic. Among three major stock indices as reported in Table 10.1, the growth and return of small-cap companies are a worse performer as compared to mid-cap and BSE Sensex, whereas, the BSE Sensex stock price is more volatile as compared to mid-cap and small-cap stock indices. Nevertheless amongst the sectors, realty, metal, Bankex and auto are a worse

Table 10.2: Trends of Growth, Stock Returns and Volatility of Stock Prices, and FII flows during the phase of Post Demonetisation, Post GST and COVID-19 pandemic

Variables	COVID-19 Pandemic				Post Demonetisation			Post GST		
	Growth	Return	Volatility	Correlation	Growth	Return	Volatility	Growth	Return	Volatility
Stock Indices										
SENSEX	-22.65	-0.54	2.77	0.15	12.07	0.08	0.51	31.96	0.05	0.59
Mid-cap	-24.15	-0.59	2.13	0.22	12.99	0.08	0.67	6.38	0.01	0.72
Small-cap	-25.96	-0.63	1.99	0.27	18.07	0.11	0.70	-4.70	-0.01	0.70
BSE100	-22.78	-0.55	2.65	0.16	12.01	0.08	0.55	23.29	0.04	0.60
BSE200	-22.76	-0.55	2.58	0.16	12.34	0.08	0.55	22.14	0.03	0.60
BSE500	-23.24	-0.56	2.53	0.17	13.16	0.08	0.55	19.36	0.03	0.60
Sectorial Indices										
Auto	-30.95	-0.77	2.69	0.21	5.18	0.04	0.81	-22.16	-0.04	0.90
Bankex	-33.17	-0.84	3.29	0.16	17.66	0.11	0.77	33.87	0.05	0.82
Cap-Goods	-30.12	-0.74	2.43	0.24	19.04	0.11	0.74	2.82	0.01	0.85
Con-dura	-21.22	-0.49	2.25	0.11	28.33	0.16	0.91	60.88	0.08	0.84
FMCG	-7.76	-0.20	2.25	0.06	21.40	0.12	0.74	10.33	0.02	0.65
Healthcare	5.76	0.09	1.96	0.17	-6.93	-0.05	0.74	0.45	0.00	0.78
IT	-19.16	-0.46	2.45	0.13	-0.15	0.00	0.83	63.96	0.08	0.81
Metal	-34.63	-0.90	3.27	0.16	8.40	0.05	1.16	-14.95	-0.02	1.27
Oil and Gas	-20.67	-0.51	2.62	0.18	10.91	0.07	0.76	9.57	0.02	0.93
Power	-21.47	-0.50	2.18	0.26	11.92	0.07	0.74	-12.93	-0.02	0.77
PSU	-28.13	-0.70	2.58	0.21	5.11	0.03	0.74	-17.41	-0.03	0.92
Realty	-41.76	-1.11	2.73	0.23	38.45	0.21	1.45	21.17	0.03	1.22
TECK	-17.87	-0.42	2.40	0.12	1.94	0.01	0.74	44.66	0.06	0.73
FII Flows										
Aggregate Flows										
Purchases	9.0	-	49.46	-0.10	11.99	-	61.47	10.6	-	57.34
Sales	7.9	-	45.10	-0.16	9.27	-	49.80	9.7	-	65.92
Net	-21.3	-	492.96	0.10	-27.20	-	1081.92	-163.8	-	5394.32
Equity						-				
Purchases	13.6	-	63.15	0.07	12.16	-	61.47	12.2	-	63.78
Sales	8.0	-	47.09	-0.05	9.38	-	50.97	9.5	-	58.88
Net	-69.9	-	1997.07	0.13	-587.07	-	8503.84	171.1	-	6845.09
Debt										
Purchases	65.9	-	209.65	-0.27	86.15	-	408.56	-	-	-
Sales	39.9	-	142.78	-0.26	65.16	-	303.68	313.7	-	6197.72
Net	42.7	-	807.85	0.07	1252.35	-	15856.43	148.2	-	5943.26

Source: Mishra, Rath and Dash (2020)

performer in terms of stock returns as well as the growth of the stock price. Since the demand for healthcare products and services are important during this COVID-19 pandemic, which could perhaps reflect in showing a positive stock price returns with low volatility. The second-best performer during this ongoing pandemic is Fast-Moving Consumer Goods (FMCG) companies, which also show a relatively less affected sector as compared to other sectors. The net FII flows figures demonstrate a negative growth during all three phases, but this indicator is less volatile in COVID-19 phase as compared to the other two events. This implies that although the average gross sales are higher than average gross purchase, growth of FII outflows from Indian financial market is still lower in COVID-19 phase as compared to post demonetisation and post-implementation of GST phases (Mishra, Rath and Dash, 2020).

Check Your Progress 2

- 1) Why is a developed stock market considered as a barometer of economic growth?

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- 2) What is market capitalisation ratio?

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- 3) Explain the structure of Indian Stock Market. What are the various reforms brought through these years in this market?

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- 4) How has the Indian Stock market been performing during the times of GST implementation and demonetisation?

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10.5 EQUITY DERIVATIVES IN INDIA

A derivative security is a financial contract whose value is derived from the value of something else, such as a stock price, a commodity price, an exchange rate, an interest rate, or even an index of prices. Derivatives may be traded for a variety of reasons. A derivative enables a trader to hedge some pre-existing risk by taking positions in derivatives markets that offset potential losses in the underlying or spot market. In India, most derivatives users describe themselves as hedgers (Fitch Ratings, 2004) and Indian laws generally require that derivatives be used for hedging purposes only. Another motive for derivatives trading is speculation (i.e. taking positions to profit from anticipated price movements). In practice, it may be difficult to distinguish whether a particular trade was for hedging or speculation, and active markets require the participation of both hedgers and speculators. The third type of trader, called arbitrageurs, profit from discrepancies in the relationship of spot and derivatives prices, and thereby help to keep markets efficient. Equity derivatives trading started in India in June 2000, after a regulatory process which stretched over more than four years. In July 2001, the equity spot market moved to rolling settlement. India's experience with the launch of the equity derivatives market has been extremely positive, by world standards. NSE is now one of the prominent exchanges amongst all emerging markets, in terms of equity derivatives turnover. There is an increasing sense that the derivatives market is playing a major role in shaping price discovery.

10.5.1 Exchange-Traded and Over-the-Counter Derivative Instruments

OTC (over-the-counter) contracts, such as forwards and swaps, are bilaterally negotiated between two parties. The terms of an OTC contract are flexible and are often customised to fit the specific requirements of the user. OTC contracts have substantial credit risk, which is the risk of the counterparty that owes money defaults on the payment. In India, OTC derivatives are generally prohibited with some exceptions: those that are specifically allowed by the Reserve Bank of India (RBI) or, in the case of commodities (which are regulated by the Forward Markets Commission), those that trade informally in "hawala" or forwards markets. An exchange-traded contract, such as a futures contract, has a

standardised format that specifies the underlying asset to be delivered, the size of the contract, and the logistics of delivery. They trade on organised exchanges with prices determined by the interaction of many buyers and sellers. In India, two exchanges offer derivatives trading: the Bombay Stock Exchange (BSE) and the National Stock Exchange (NSE). A snapshot of turnover in BSE is presented in Table 10.3. However, NSE now accounts for virtually all exchange-traded derivatives in India, accounting for more than 99 per cent of volume in 2003-2004. Contract performance is guaranteed by a clearinghouse, which is a wholly-owned subsidiary of the NSE. Margin requirements and daily marking-to-market of futures positions substantially reduce the credit risk of exchange-traded contracts, relative to OTC contracts (Asani Sarkar, 2006).

In the exchange-traded market, the biggest success story has been derivatives on equity products. Index futures were introduced on June 12, 2000, followed by index options on June 4, 2001, and options and futures on individual securities in July 2001 and November 2001, respectively. As of 2005, the NSE trades futures and options on 118 individual stocks and 3 stock indices. The NSE currently provides trading in futures and options contracts on indices such as Nifty 50 index, Nifty IT index, Nifty Bank index, Nifty Midcap 50 index, Nifty Infrastructure index, Nifty PSE index, and Nifty CPSE. All these derivative contracts are settled by cash payment and do not involve physical delivery of the underlying product. Derivatives on stock indexes and individual stocks have grown rapidly since inception. In particular, single stock futures have become hugely popular, accounting for about half of NSE's traded value in October 2005. NSE has the highest volume (i.e. the number of contracts traded) in the single stock futures globally, enabling it to rank 16 among world exchanges in the first half of 2005. Single stock options are less popular than futures. Index futures are increasingly popular and accounted for close to 40 per cent of traded value in October 2005. NSE launched interest rate futures in June 2003 but, in contrast to equity derivatives, there has been little trading in them. One problem with these instruments was faulty contract specifications, resulting in the underlying interest rate deviating erratically from the reference rate used by market participants. Institutional investors have preferred to trade in the OTC markets, where instruments such as interest rate swaps and forward rate agreements are thriving. As interest rates in India have fallen, companies have swapped their fixed rate borrowings into floating rates to reduce funding costs. However, it may be noted that the detailed discussion of Equity Derivative instruments is beyond the scope of this unit. Table 10.4 presents the turnover of stock options in both BSE and NSE in panel A and B respectively. The NSE witnessed a domestic market share of 84 per cent in the cash market segment with 100 per cent market share in the equity derivatives segment during fiscal 2017 (Annual Report of NSE, 2017).

Table10.3: Turnover (BSE) in Indian Equity Derivatives Market

Period	No. of Trading Days	Index Future Turnover (₹ crore)	Stock Future Turnover (₹ crore)	Index Option	
				Call	Put
				Notional Turnover (₹ crore)	Notional Turnover (₹ crore)
2010-11	254	154.08	0.00	0.00	0.25
2011-12	249	178,448.82	10,215.70	200,089.59	418,252.79
2012-13	249	122,373.70	3,418.27	3,230,231.91	3,797,249.17
2013-14	251	63,493.81	54,609.21	5,705,316.57	3,349,884.08
2014-15	243	48,632.28	9,794.23	10,112,605.17	10,016,621.27
2015-16	247	13,097.14	1,349.66	2,560,540.64	1,825,708.15
2016-17	248	2,266.96	203.07	1,254.89	3,214.45
2017-18	246	3,217.54	36.73	5.92	2.28
2018-19	248	39.16	17.78	1,308.50	884.63
2019-20*	183	8,362.57	150.63	13,190.06	6,432.51

Source: BSE and SEBI

Note: * indicates upto 31st Dec 2019, Turnover in notional value.**Table 10. 4: Turnover in Indian Equity Derivatives Market (BSE & NSE)**

Period	Panel-A Stock Options (BSE)						Total	
	Call			Put			No. of Contracts	Notional Turnover (₹ crore)
	No. of Contracts	Premium Turnover (₹ crore)	Notional Turnover (₹ crore)	No. of Contracts	Premium Turnover (₹ crore)	Notional Turnover (₹ crore)		
2010-11	0	0.00	0.00	0	0.00	0.00	5,623	154.33
2011-12	39,848	0.00	1,277.28	7,657	0.00	191.82	3,22,22,825	808,476.00
2012-13	1,78,313	0.00	5,185.77	2,09,557	0.00	5,059.86	26,24,43,366	7,163,518.70
2013-14	6,67,365	0.00	22,185.51	8,77,355	0.00	23,945.21	30,19,42,441	9,219,434.44
2014-15	30,10,092	0.00	93,854.42	27,00,450	0.00	81,233.89	50,54,78,869	20,362,741.27
2015-16	10,09,439	374.25	31,904.12	14,13,452	351.32	42,408.54	10,62,09,394	4,475,008.26
2016-17	0	0.00	0.00	0	0.00	0.00	1,23,538	6,939.37
2017-18	3	0.00	0.18	0	0.00	0.00	44,701	3,262.65
2018-19	2	0.00	0.08	0	0.00	0.00	31,167	2,250.14
2019-20*	8,472	8.96	626.03	7,876	7.58	583.24	2,90,395	29,345.04

Panel-B

	Period Stock Options (NSE)						Total	
	Call			Put			No. of Contracts	Notional Turnover (₹ crore)
	No. of Contracts	Notional Turnover (₹ crore)	Notional Turnover (₹ crore)	No. of Contracts	Premium Turnover (₹ crore)	Notional Turnover (₹ crore)		
2010-11	2,42,73,560	15,710	7,77,109	82,34,833	4,765	2,53,235	103,42,12,062	2,92,48,221
2011-12	2,45,65,283	13,331	6,71,770	1,19,29,088	6,282	3,05,261	120,50,45,464	3,13,49,732
2012-13	4,24,99,219	22,583	13,02,779	2,42,78,974	11,705	6,97,648	113,14,67,418	3,15,33,004
2013-14	5,03,00,025	29,812	15,43,895	2,98,74,406	16,617	8,65,594	128,44,24,321	3,82,11,408
2014-15	6,12,04,473	43,345	22,43,382	3,02,74,736	18,387	10,39,170	183,70,41,131	5,56,06,453
2015-16	6,53,22,962	39,014	23,25,030	3,49,76,212	22,380	11,63,144	209,86,10,395	6,48,25,834
2016-17	6,12,05,774	65,782	41,47,488	3,09,00,238	29,788	19,59,998	139,97,46,129	9,43,70,302
2017-18	8,62,82,094	1,02,165	67,28,007	4,01,29,282	46,053	29,27,002	191,38,78,548	16,49,84,859
2018-19	12,35,10,308	1,28,529	85,17,920	6,34,76,234	71,481	40,64,454	316,48,02,420	23,76,00,705
2019-20*	9,44,82,765	96,372	58,15,639	5,46,53,714	61,374	31,37,545	367,40,28,197	24,64,27,009

Source: BSE, NSE and SEBI

Note: * indicates upto 31st Dec 2019, Turnover in notional value.

Table 10.5: Turnovers in Index Derivatives

Period	(Per cent)					
	BSE		NSE		MSEI**	
	Index Futures	Index Options	Index Futures	Index Options	Index Future	Index Option
2010-2011	100.00	-	19.17	80.83	NA	NA
2011-2012	22.38	77.62	13.61	86.39	NA	NA
2012-2013	1.76	98.24	9.99	90.01	NA	NA
2013-2014	0.75	99.25	10.00	90.00	45.13	54.87
2014-2015	0.24	99.76	9.33	90.67	100.00	-
2015-2016	0.30	99.70	8.52	91.48	-	-
2016-2017	33.65	66.35	5.62	94.38	-	-
2017-2018	99.75	0.25	3.44	96.56	-	-
2018-2019	1.75	98.25	2.67	97.33	-	-
2019-2020*	85.77	14.23	2.06	97.94	-	-

Source: BSE, NSE, MSEI and SEBI

Note: * indicates upto 31st Dec 2019, ** Trading in equity derivatives segment for Index future and option has commenced from May 15, 2013

Table 10.5 reports the turnovers in index futures and index options in three stock exchange such as Bombay Stock Exchange (BSE), National Stock Exchange (NSE), Metropolitan Stock Exchange of India Limited (MSEI). The turnover per cent in both BSE and NSE for the index futures are affected in the year of